

Basic Game Logic Description (Chess)

This project implements a turn-based chess game with a clear separation between core logic and UI.

The game starts in ACTIVE state and ends in one of terminal states: WHITE_WON, BLACK_WON, or DRAW.

Board generation and initial setup

- At game start, the controller creates a standard 8×8 board and places all pieces in initial chess positions.
- Create board with 64 squares (row, column, color).
- Create two players (WHITE, BLACK).
- Place pieces:
 - White major pieces on rank 1, white pawns on rank 2.
 - Black major pieces on rank 8, black pawns on rank 7.
- Set currentPlayer = WHITE.
- Set gameState = ACTIVE.

Player move flow

A move is processed as one turn cycle:

1. Player selects a piece.
2. System verifies the piece belongs to the current player.
3. System generates candidate moves for that piece (piece-specific movement rules).
4. System filters candidates to legal moves
5. If target square matches a legal move, execute move:
 1. update piece position,
 2. handle capture (if any),
 3. handle special moves (castling, en passant, promotion).
6. Switch current player.
7. Recompute game state.

If move is invalid, state remains unchanged and player must try again.

Game state validation after each move

After every valid move, the controller evaluates game status:

- Win condition (WHITE_WON / BLACK_WON):

- opponent king is in check and
- opponent has no legal move.
- Draw condition (DRAW):
 - stalemate (not in check, but no legal moves),
 - insufficient material,
 - threefold repetition.
- Otherwise:
 - state stays ACTIVE.

Piece/state transition logic during move

During move execution, the following object state changes happen:

1. source square piece reference becomes null,
2. destination square receives moved piece,
3. moved piece updates its internal square,
4. captured piece is removed from opponent piece list,
5. piece-specific flags may update (firstMove for king/rook/pawn),
6. game state may transition from ACTIVE to terminal state.

UI interaction responsibility

UI does not decide chess rules.

UI only:

- displays board and messages,
- reads user input,
- sends commands to controller,
- re-renders board after controller response.

Controller is the single source of truth for board, turn, and game state.